

Rectangular matrix multiplication from shared-leg entropy

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Pathway

October 7, 2026

Abstract

In this note, we extend the matrix-multiplication analysis of [6] to rectangular products and prove

$$\omega(1, k, 1) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + 1/(4k), & k \geq \frac{1}{2}. \end{cases}$$

In particular, $\omega(1, \frac{1}{2}, 1) = 2$ and the dual exponent satisfies $\alpha \geq \frac{1}{2}$. We use the shared-leg entropy inequality and polynomial-multiplication degenerations of OpenAI, retaining two-leg symmetry and the orientation of each sector. Logarithmic averaging produces homogeneous auxiliary profiles. Their powered versions have a common asymptotic slope, and bounding their intercepts gives the spectral constraint $b \leq 4a(1 - a)$. This yields the rectangular curve by tensor-spectrum duality. As an application, Zwick’s algorithm [10] for all-pairs shortest paths in directed unweighted graphs runs in $O(n^{2.5})$ time. Combining the rectangular bound with the $(\min, +)$ -product improvement of Alman and Vassilevska Williams [1] further gives $O(n^{2.4999})$ running time.

1 Introduction

The motivation for this note is to gather, in one place, useful bounds and algorithmic consequences of the groundbreaking matrix-multiplication result of [6], with a focus on rectangular products.

Complexity convention. Throughout this note, all complexity bounds suppress a $+o(1)$ term in the exponent: $O(n^c)$ denotes $n^{c+o(1)}$. All shape parameters are fixed.

Rectangular matrix multiplication is a basic ingredient in algorithms for all-pairs problems, including shortest paths, lowest common ancestors, and Hamming distance. Its exponents depend on the shape of the product; an improvement for square matrices need not give a comparable improvement for every rectangular shape. OpenAI [6] proved the square bound $\omega \leq 9/4$ using a shared-leg entropy inequality and degenerations of polynomial-multiplication tensors. We extend that analysis to rectangular products by retaining the orientation information lost under full leg symmetrization.

For fixed nonnegative real u, v, w , let $\omega(u, v, w)$ be the infimum of the exponents e for which multiplying an $\lceil n^u \rceil \times \lceil n^v \rceil$ matrix by a $\lceil n^v \rceil \times \lceil n^w \rceil$ matrix uses $O(n^e)$ arithmetic operations over $\mathbb{C}$, with the convention above. Write $\omega(k) = \omega(1, k, 1)$ and $\alpha = \sup\{k \geq 0 : \omega(k) = 2\}$.

*This note, its proofs, analysis and writeup were obtained fully through OpenAI Codex (GPT-6.1). The author has read and analyzed this note in full and takes full responsibility for its contents.

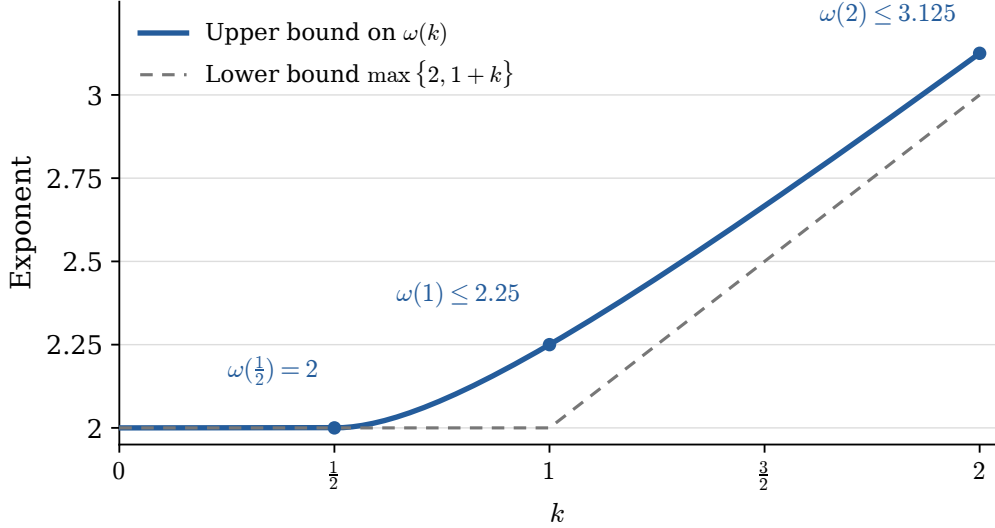


Figure 1: The upper bound on $\omega(k)$ from Theorem 1.1 for $0 \leq k \leq 2$, together with the size lower bound $\max\{2, 1+k\}$. The bounds coincide for $0 \leq k \leq \frac{1}{2}$.

Theorem 1.1. *For every fixed real $k \geq 0$,*

$$\omega(k) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases} \quad (1)$$

Consequently $\omega(k) = 2$ for $0 \leq k \leq \frac{1}{2}$, $\alpha \geq \frac{1}{2}$, and $\omega(2) \leq 3.125$.

Theorem 1.1 gives exponent two for inner dimension up to $n^{0.5}$, and an upper bound within $1/(4k)$ of the input-size lower bound $1+k$ for $k \geq 1$.

Our proof bounds the dot-product parameters of every tensor character. After averaging over the interchange of two legs, these parameters have the form (a_0, a_0, b_0) , and the rectangular objective is $2a_0 + kb_0$. We prove the parabolic constraint $b_0 \leq 4a_0(1 - a_0)$.

Relation to the preceding analysis. The tensor constructions and shared-leg entropy inequality are taken directly from [6]. Lemma 2.1 restates its Corollary 3.2 and the constructions in the proofs of its Lemmas 4.1 and 4.2, with oriented notation. The partial symmetrization in Section 2 adapts its six-permutation geometric averaging to two permutations. The powered concavity and coupled tripling inequalities in Lemma 2.2 are direct consequences of those same constructions, retaining the two distinct orientations.

The additional analytic arguments relative to that paper are the simultaneous logarithmic homogenization of the two profiles (Lemma 3.1) and their common-slope and intercept analysis (Lemma 4.1), which yields the parabolic parameter bound (Proposition 4.2). The underlying tools, including geometric averaging, compactness, and concavity, are standard. These arguments extend the original use of concavity and tripling to constrain two parameters. Once this constraint is established, Section 5 obtains the curve by elementary optimization and standard tensor-spectrum duality; Section 6 optimizes the permuted constraints for three unequal dimensions; Section 7 substitutes it into known algorithmic tradeoffs. Thus the contribution is the rectangular bound and the coupled-profile analysis leading to it; the tensor constructions and the application algorithms are inherited.

2 Tensor characters and polynomial multiplication

We view a tensor as a trilinear form in three groups of variables, called its *legs*. A *restriction* applies linear substitutions independently on the three legs; a *degeneration* is a limit of such restrictions.

A *tensor character* assigns each tensor T a nonnegative real number $\lambda(T)$, with

$$\lambda(xyz) = 1, \quad \lambda(S \oplus T) = \lambda(S) + \lambda(T), \quad \lambda(S \otimes T) = \lambda(S)\lambda(T).$$

Here $\oplus$ uses disjoint variable groups and $\otimes$ is the Kronecker product. A character is also monotone: if T restricts or degenerates to S , then $\lambda(S) \leq \lambda(T)$. The collection of these characters is the *tensor spectrum*. For the oriented dot-product tensors

$$B_X(q) = x \sum_{j=1}^q y_j z_j,$$

and its leg permutations $B_Y(q), B_Z(q)$, write

$$\lambda(B_i(q)) = q^{p_i}, \quad 0 \leq p_i \leq 1. \quad (2)$$

The *rank* of a tensor is the minimum number of products of three linear forms, one in each leg, whose sum equals that tensor. Characters are positive on nonzero tensors and bounded above by rank. The matrix-multiplication tensor $\langle A, B, C \rangle = \sum_{u,v,w} x_{uv} y_{vw} z_{wu}$ has character value

$$\lambda(\langle A, B, C \rangle) = A^{p_Y} B^{p_Z} C^{p_X}, \quad (3)$$

by its factorization into three oriented dot products. Define the asymptotic rank by $\widetilde{\text{rank}}(T) = \lim_{m \rightarrow \infty} \text{rank}(T^{\otimes m})^{1/m}$. We use the standard duality

$$\widetilde{\text{rank}}(T) = \max_{\lambda} \lambda(T), \quad (4)$$

as stated, for example, in [2, Section 4.1, p. 18].

For distinct tensor legs i, j , put the polynomial-multiplication tensor

$$C_{ij}(r, s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_{\ell}^{(j)} x_{h+\ell}^{(k)}$$

on input legs i, j and the remaining output leg k . Its rank is $r + s - 1$ over $\mathbb{C}$: interpolation at $r + s - 1$ distinct points gives the upper bound, and the output flattening gives the lower bound. A smaller pair of input lengths gives a restriction.

The following results are Corollary 3.2 and the constructions in the proofs of Lemmas 4.1 and 4.2 of OpenAI [6], pp. 5–8.

Lemma 2.1 (OpenAI). *The following statements hold for every leg permutation.*

1. *If a tensor has matched sectors T_h sharing leg i , with the sectors disjoint on the other two legs, then for every probability vector q ,*

$$\lambda(T) \geq \exp(p_i H(q)) \prod_h \lambda(T_h)^{q_h}, \quad H(q) = - \sum_h q_h \log q_h. \quad (5)$$

The same inequality applies to a degeneration onto such sectors.

2. *The determinant filtration of $C_{ij}(r, s) \otimes B_i(2)$, for $s \geq 2$, yields two matched sectors $C_{ij}(r, s+1)$ and $C_{ij}(r, s-1)$ sharing leg i .*
3. *For $r, s \geq 1$, $C_{ij}(r, 3s+r-1)$ degenerates to three matched sectors sharing leg i : two copies of $C_{ij}(r, s)$ and one copy of $C_{ik}(r, s)$, where k is the remaining leg.*

2.1 Partial symmetrization

We adapt the geometric averaging in [6, Section 4], retaining only the interchange of X, Y . The use of a common probability vector before geometric averaging is also taken from that argument.

Fix a character λ . Let τ exchange X, Y , and define

$$M(T) = \sqrt{\lambda(T)\lambda(T^\tau)}, \quad a_0 = \frac{p_X + p_Y}{2}, \quad b_0 = p_Z. \quad (6)$$

The function M is positive, multiplicative, invariant under τ , monotone under restriction and degeneration, and bounded above by rank. Its dot-product parameters are $(\bar{p}_X, \bar{p}_Y, \bar{p}_Z) = (a_0, a_0, b_0)$. It need not be additive and is not treated as a character.

Apply (5) to both factors in M with the same probability vector and take their geometric mean. For sectors with common leg i , this gives the same entropy inequality for M , using the averaged parameter $\bar{p}_i$. For $\bar{p}_i > 0$, the entropy variational identity gives

$$M(T)^{1/\bar{p}_i} \geq \sum_h M(T_h)^{1/\bar{p}_i}. \quad (7)$$

The probability vector is optimized after averaging the two character inequalities.

2.2 Discrete profiles

A *profile* here is a function of the two input lengths. Set

$$V_d(r, s) = M(C_{XY}(r, s)), \quad U_d(r, s) = M(C_{XZ}(r, s)). \quad (8)$$

The first profile is symmetric in r, s ; X/Y symmetry identifies $M(C_{XZ}(r, s))$ and $M(C_{YZ}(r, s))$. Both profiles are nondecreasing in each coordinate and satisfy

$$1 \leq V_d(r, s), U_d(r, s) \leq r + s - 1. \quad (9)$$

Lemma 2.2 (Powered concavity and tripling). *If $a_0 > 0$, V_d^{1/a_0} is concave separately in r, s , and U_d^{1/a_0} is concave in s . If $b_0 > 0$, U_d^{1/b_0} is concave in r . A zero parameter makes the corresponding profile constant in that coordinate. For positive denominators, the tripling inequalities are*

$$V_d(r, 3s + r - 1)^{1/a_0} \geq 2V_d(r, s)^{1/a_0} + U_d(r, s)^{1/a_0}, \quad (10)$$

$$U_d(r, 3s + r - 1)^{1/a_0} \geq 2U_d(r, s)^{1/a_0} + V_d(r, s)^{1/a_0}, \quad (11)$$

$$U_d(3r + s - 1, s)^{1/b_0} \geq 3U_d(r, s)^{1/b_0} \quad (b_0 > 0). \quad (12)$$

Proof. The determinant filtration and (7) give

$$2M(C_{ij}(r, s))^{1/\bar{p}_i} \geq M(C_{ij}(r, s+1))^{1/\bar{p}_i} + M(C_{ij}(r, s-1))^{1/\bar{p}_i}.$$

Exchanging the inputs gives the powered concavity in the other length. If $\bar{p}_i = 0$, choosing a probability vector supported on the larger-length sector in (5) gives $M(C_{ij}(r, s)) \geq M(C_{ij}(r, s+1))$. Restriction monotonicity gives the reverse inequality, proving constancy without division by zero.

The three-sector input and (7) give (10)–(11). For (12), view $C_{XZ}(r, s)$ as $C_{ZX}(s, r)$. Its middle sector is $C_{ZY}(s, r)$, which has the same M value by X/Y symmetry. All three sectors therefore have the same value in this orientation. $\square$

If $a_0 = 0$, V_d is constant in both coordinates and equals one. However, $C_{XY}(n, n)$ restricts to $B_Z(n)$ by retaining output coefficient $n - 1$, so $V_d(n, n) \geq n^{b_0}$ and $b_0 = 0$. The desired parameter bound is immediate in this case. Assume $a_0 > 0$ below.

Lemma 2.3 (Coordinate comparisons and bounded orientation ratio). *If g is a positive nondecreasing concave sequence, then for $m' \geq m \geq 2$,*

$$\frac{g(m')}{g(m)} \leq \frac{m' - 1}{m - 1}. \quad (13)$$

For $n \geq 2$,

$$\left(2 + \frac{2}{n - 1}\right)^{-1} \leq \left(\frac{U_d(n, n)}{V_d(n, n)}\right)^{1/a_0} \leq 2 + \frac{2}{n - 1}. \quad (14)$$

Proof. The secant inequality gives $g(m') \leq g(m) + (m' - m)(g(m) - g(1))/(m - 1)$, proving (13). Apply it to the powered profiles with $r = n$, $m = n$, $m' = 4n - 1$ in (10). The upper factor is $(4n - 2)/(n - 1) = 4 + 2/(n - 1)$; subtracting the two outer-sector values gives the upper bound in (14). Use (11) for the reciprocal bound. $\square$

3 Homogeneous auxiliary profiles

The proof of [6, Lemma 5.1] uses a discrete recurrence along the diagonal of one fully symmetrized profile. We construct homogeneous auxiliary profiles from the discrete values in (8), preserving powered concavity and the tripling inequalities.

Lemma 3.1 (Logarithmic homogenization). *There exist positive locally Lipschitz profiles V, U on $(0, \infty)^2$ and a number*

$$\max(a_0, b_0) \leq \delta \leq 1 \quad (15)$$

such that both profiles are homogeneous of degree δ , $V(1, 1) = 1$, V is symmetric, and both are nondecreasing. They inherit all powered concavities in Lemma 2.2 and its tripling inequalities with $3s + r$ and $3r + s$ in place of the integer shifts.

Proof. Define, for real t ,

$$L_V(r, s; t) = \log V_d(\lceil re^t \rceil, \lceil se^t \rceil), \quad L_U(r, s; t) = \log U_d(\lceil re^t \rceil, \lceil se^t \rceil),$$

and average using one common normalization:

$$v_T(r, s) = \frac{1}{T} \int_0^T (L_V(r, s; t) - L_V(1, 1; t)) dt, \quad (16)$$

$$u_T(r, s) = \frac{1}{T} \int_0^T (L_U(r, s; t) - L_U(1, 1; t)) dt. \quad (17)$$

Applying (13) after the relevant power bounds a coordinate change in a log profile by its parameter times the corresponding log-length change, with $O(e^{-t})$ rounding errors on a fixed compact positive rectangle. A coordinate with zero parameter is constant. Together with (14), these comparisons bound the two integrands in (16)–(17) on each compact rectangle, uniformly in $t \geq 0$. They also bound an averaged difference between two nearby points by a constant times their log-coordinate distance plus $O(1/T)$. Interpolating integer log profiles changes the averages by $O(1/T)$ and gives uniformly Lipschitz approximations. Applying Arzelà–Ascoli on expanding compact rectangles and taking a diagonal subsequence gives locally uniform

limits v, u along a common sequence $T_j \rightarrow \infty$.¹ Passing to a further subsequence, the bounded scalars also satisfy

$$\delta_{T_j} = \frac{L_V(1, 1; T_j)}{T_j} \rightarrow \delta.$$

All comparison constants here may depend on the fixed character.

The rank bound (9) gives $\delta \leq 1$. The tensor $C_{XY}(n, n)$ restricts to $B_X(n), B_Y(n), B_Z(n)$: retain the first-input constant coefficient, the second-input constant coefficient, or output coefficient $n - 1$, respectively. Thus $V_d(n, n) \geq n^{\max(a_0, b_0)}$, proving (15). In particular $\delta > 0$.

Fix $c > 0$ and put $d = \log c$. Replacing (r, s) by (cr, cs) shifts the first term of either integral by d in its t argument. The difference is the sum of two endpoint integrals. On the upper interval, for bounded w ,

$$L_V(r, s; T + w) = L_V(1, 1; T) + O_{r,s,d}(1),$$

uniformly; the same holds for L_U by (14). The lower endpoint contributes $O(1)$. Dividing by T gives

$$v(cr, cs) - v(r, s) = \delta \log c, \quad u(cr, cs) - u(r, s) = \delta \log c.$$

This also handles $d < 0$, since L_V, L_U are defined for all real t . Consequently $V = e^v, U = e^u$ have degree δ and $V(1, 1) = 1$.

It remains to justify the inequalities in the limit. The geometric-mean functional $G_T(h) = \exp(T^{-1} \int_0^T \log h(t) dt)$ is concave and homogeneous on positive functions. In particular

$$h_t \geq 2g_t + j_t \implies G_T(h) \geq 2G_T(g) + G_T(j). \quad (18)$$

For a direct proof, use the entropy variational identity for $\log(2g_t + j_t)$ with a single probability vector, integrate, and optimize that vector after integration. The same argument with two weighted terms shows that geometric averaging preserves concavity. An integrand-dependent positive multiplier common to all profiles does not affect these statements.

Integer powered profiles have concave piecewise linear interpolations in each applicable coordinate. Rounding a length changes a log profile by $O(e^{-t})$ on compact positive rectangles by (13). Thus their concavity inequalities, after averaging, have multiplicative errors $\exp(O(1/T))$. Likewise, $3\lceil se^t \rceil + \lceil re^t \rceil - 1$ and $\lceil (3s + r)e^t \rceil$ differ by a bounded integer; changing between them changes the log value by $O(e^{-t})$. Apply (18) to (10)–(12) with the common normalization in (16)–(17). The errors vanish as $T \rightarrow \infty$, giving the exact continuum tripling inequalities. Symmetry and monotonicity also pass to the limit. $\square$

Replace these profiles by their powers $1/\delta$, and set

$$a = a_0/\delta, \quad b = b_0/\delta. \quad (19)$$

We reuse the names V, U for the resulting degree-one profiles. They have $0 < a \leq 1, 0 \leq b \leq 1$, and $V(1, 1) = 1$. Powered concavity and tripling are unchanged, since $(V^{1/\delta})^{1/a} = V^{1/a_0}$, and similarly for the other powers.

¹On each rectangle $K_j = [1/j, j]^2$, the interpolated profiles are uniformly bounded and have a common Lipschitz bound, so Arzelà–Ascoli gives a subsequence on which both converge uniformly. Choose nested subsequences for $K_1, K_2, \dots$, then take the j -th term of the j -th subsequence. The resulting sequence converges uniformly on every compact subset of $(0, \infty)^2$; the $O(1/T)$ interpolation error gives the same limits for the original profiles.

4 Common slopes and bounded intercepts

Put

$$P(r, s) = V(r, s)^{1/a}, \quad Q(r, s) = U(r, s)^{1/a}, \quad D = 1/a. \quad (20)$$

Both profiles are homogeneous of degree D and concave and nondecreasing in s . The profile P is symmetric and concave in r . Positive separate concavity gives $P(2r, 2s) \leq 4P(r, s)$, so $D \leq 2$ and $a \geq \frac{1}{2}$. Here one extends each positive nondecreasing concave one-variable profile to its nonnegative limit at zero before applying the midpoint inequality.

Lemma 4.1 (Common slope and intercepts). *For $r = 1$, write $P(s) = P(1, s)$ and $Q(s) = Q(1, s)$. There is a common slope $L > 0$ such that*

$$\lim_{s \rightarrow \infty} \frac{P(s)}{s} = \lim_{s \rightarrow \infty} \frac{Q(s)}{s} = L.$$

The limits

$$C_P = \lim_{s \rightarrow \infty} (P(s) - Ls), \quad C_Q = \lim_{s \rightarrow \infty} (Q(s) - Ls)$$

exist, are nonnegative and finite, and satisfy

$$C_P + C_Q \leq L, \quad C_P \geq 1 - L, \quad C_Q \leq 2L - 1, \quad L \leq \frac{1}{2a}. \quad (21)$$

Proof. Positive nondecreasing concavity gives finite nonnegative limits $L_P = \lim P(s)/s$, $L_Q = \lim Q(s)/s$. Divide the two coupled tripling inequalities

$$P(3s + 1) \geq 2P(s) + Q(s), \quad Q(3s + 1) \geq 2Q(s) + P(s)$$

by s and pass to infinity. They give $L_P \geq L_Q$ and $L_Q \geq L_P$, so the slopes agree.

Their sum $R = P + Q$ satisfies $R(3s + 1) \geq 3R(s)$. Iterate at $s_j = 3^j(s + \frac{1}{2}) - \frac{1}{2}$ and use $R(s_j)/s_j \rightarrow 2L$ to get

$$R(s) \leq L(2s + 1). \quad (22)$$

Since $R(s) > 0$, this proves $L > 0$.

The derivatives of a concave profile decrease to its asymptotic slope. Hence $P(s) - Ls$ and $Q(s) - Ls$ are nondecreasing. Each is nonnegative by comparison with its nonnegative limit at zero. Equation (22) bounds their sum by L , so their finite limits exist and $C_P + C_Q \leq L$. As $P(1) = 1$, $C_P \geq 1 - L$, and therefore $C_Q \leq 2L - 1$.

Symmetry and homogeneity give $P(1, s) = s^D P(1, 1/s)$. Taking one-sided derivatives at $s = 1$ yields $P'_+(1) + P'_-(1) = D$. Concavity implies $P'_+(1) \leq D/2$. The limiting slope is at most $P'_+(1)$, proving $L \leq 1/(2a)$. No differentiability at $s = 1$ is needed. $\square$

Proposition 4.2 (Parabolic parameter cap). *The normalized parameters satisfy $b \leq 4a(1 - a)$.*

Proof. At differentiability points, $Q'_s(1, s) \geq L$ and $Q(1, s) \leq Ls + C_Q \leq Ls + 2L - 1$. Thus

$$\partial_s \log U(1, s) = a \frac{Q'_s(1, s)}{Q(1, s)} \geq \frac{aL}{Ls + 2L - 1} \geq \frac{a}{s + 2(1 - a)}. \quad (23)$$

The denominators are positive: $L > 0$ and $2L - 1 \geq C_Q \geq 0$. The last inequality uses $2 - 1/L \leq 2 - 2a$. By degree-one homogeneity,

$$\partial_s \log U(r, s) \geq \frac{a}{s + 2(1 - a)r}. \quad (24)$$

For $b > 0$, the continuum self-tripling inequality and concavity in r of $U^{1/b}$ give

$$\partial_r \log U(r, s) \geq \frac{2b}{2r + s}. \quad (25)$$

Indeed, the forward secant from r to $3r + s$ is at most the right derivative at r , while the increment is at least $2U(r, s)^{1/b}$. If $b = 0$, monotonicity gives (25) with zero right-hand side without taking a power.

Euler's identity for degree-one U now gives, almost everywhere,

$$1 \geq \frac{2br}{2r + s} + \frac{as}{s + 2(1 - a)r}. \quad (26)$$

The right-hand side is continuous in r/s , so the inequality holds at every positive ratio. Put $x = 2r/s$ and $A = 1 - a$; rearranging yields

$$b \leq \frac{A(1 + x)^2}{x(1 + Ax)} \quad (x > 0). \quad (27)$$

For $\frac{1}{2} < a < 1$, choose $x = 1/(2a - 1)$, which gives $b \leq 4a(1 - a)$. At $a = \frac{1}{2}$, let $x \rightarrow \infty$ to get $b \leq 1$. At $a = 1$, (26) gives $b = 0$ directly. $\square$

5 The rectangular exponent curve

Corollary 5.1 (Character constraint). *Every character satisfies*

$$b_0 \leq 4a_0(1 - a_0), \quad a_0 = (p_X + p_Y)/2, \quad b_0 = p_Z. \quad (28)$$

Proof. For $a_0 > 0$, Proposition 4.2 and (19) give

$$b_0 \leq 4a_0 \left(1 - \frac{a_0}{\delta}\right) \leq 4a_0(1 - a_0),$$

since $\delta \leq 1$. If $a_0 = 0$, we already proved $b_0 = 0$. $\square$

Proof of Theorem 1.1. For $\langle q, q^k, q \rangle$, (3) gives the character exponent $p_X + p_Y + kp_Z = 2a_0 + kb_0$. Corollary 5.1 therefore bounds this exponent by

$$\max_{0 \leq a \leq 1} (2a + 4ka(1 - a)).$$

For $0 \leq k \leq \frac{1}{2}$ the maximum is 2, attained at $a = 1$. For $k \geq \frac{1}{2}$ the optimizer is

$$a_* = \frac{1}{2} + \frac{1}{4k}, \quad b_* = 4a_*(1 - a_*) = 1 - \frac{1}{4k^2}, \quad (29)$$

and the value is $1 + k + 1/(4k)$. The optimization is over the parameter region defined by Corollary 5.1; it does not require a character attaining (29).

For rational $k \geq 0$, choose arbitrarily large integers q with q^k integral. The preceding bound on every character and (4) give the same upper bound on $\log_q \text{rank}(\langle q, q^k, q \rangle)$. Tensor-power rank decompositions yield algorithms with this exponent bound: choose a tensor power with rank within an arbitrary exponent slack and recurse with that fixed rectangular decomposition. Additions are covered by its rank and flattening dimensions. Padding from these base sizes to arbitrary n costs only a constant factor for each fixed algorithm.

For real k , approximate from above by rational inner exponents and pad. Equivalently, inner-dimension blocking gives $\omega(k + h) \leq \omega(k) + h$ for $h \geq 0$. Continuity of the right-hand side of (1) and the exponent convention complete the extension to all real k . The output flattening has rank n^2 , giving $\omega(k) \geq 2$ and proving equality when $k \leq \frac{1}{2}$ and the claim about α . $\square$

Some values of the bound in Theorem 1.1 are:

k	0.5	0.6	0.8	1	1.5	2
$\omega(k) \leq$	2	2.0166...	2.1125	2.25	2.6666...	3.125

The exponent convention gives an $O(n^2)$ -time algorithm at $k = \frac{1}{2}$.

6 Three unequal dimensions

Permuting the parabolic character constraint gives a bound for general $\omega(a, b, c)$ by a further elementary optimization.

Corollary 6.1. *For every fixed $a \geq b \geq c \geq 0$,*

$$\omega(a, b, c) \leq \begin{cases} a + b, & b \geq 2c, \\ a + b + \frac{(2c - b)^2}{4(a + c - b)}, & b < 2c. \end{cases} \quad (30)$$

When $b \geq 2c$, equality holds by the size lower bound.

Proof. Corollary 5.1, applied after each leg permutation, gives $p_i \leq 2(p_j + p_\ell) - (p_j + p_\ell)^2$ for distinct i, j, ℓ . Reorder the parameters so that the character objective is $ap_1 + bp_2 + cp_3$, and put $S = p_1 + p_2 + p_3$. If $S \leq 2$, this objective is at most $a + b$, since $0 \leq p_i \leq 1$. Otherwise the constraints give $S \leq 3(S - p_i) - (S - p_i)^2 \leq 9/4$. Writing $d = \sqrt{9 - 4S} \in [0, 1]$, they imply

$$\frac{3 - 2d - d^2}{4} \leq p_i \leq \frac{3 + 2d - d^2}{4}.$$

Consequently,

$$ap_1 + bp_2 + cp_3 = bS + (a - b)p_1 - (b - c)p_3 \leq \frac{3(a + b + c)}{4} + \frac{a - c}{2}d - \frac{a + c - b}{4}d^2.$$

For $b \geq 2c$, the quadratic is maximized on $[0, 1]$ at $d = 1$. For $b < 2c$, its maximum is at $d = (a - c)/(a + c - b)$; here $c > 0$ and the denominator is positive. These two maxima give (30) by tensor-spectrum duality, with the same rounding and exponent convention as in Theorem 1.1. $\square$

7 Applications

Theorem 1.1 and Corollary 6.1 give the following bounds through standard rectangular-multiplication balances. All bounds use arithmetic operations, with graph operations and symbol comparisons also charged.

1. **One lowest common ancestor per pair in a DAG.** Grandoni et al. [4, Theorem 4.2 of the published version] obtain exponent $1 + 2x$ from the balance $\omega(x) = 3x$. Theorem 1.1 permits $x = (1 + \sqrt{3})/4$, giving $O(n^{(3 + \sqrt{3})/2})$ time for finding one lowest common ancestor for every pair in an n -vertex DAG. The exponent is $(3 + \sqrt{3})/2 \approx 2.36603$.
2. **Exact all-pairs Hamming distance.** For n vectors of length n over an unrestricted alphabet, the reduction to dominance products [5] and Yuster's tradeoff [8, Proposition 2.1] give the balance $\rho = \omega(4 - \rho)$. Theorem 1.1 gives $\rho = (13 - \sqrt{7})/4 \approx 2.58856$ and running time $O(n^\rho)$ for computing every exact Hamming distance.

3. **The standard unweighted APSP balance.** Zwick’s deterministic algorithm for directed unweighted APSP [10, Section 6] has exponent $2 + \mu$, where $\omega(\mu) \leq 1 + 2\mu$. Here $\omega(\frac{1}{2}) = 2 = 1 + 2(\frac{1}{2})$, so it gives $O(n^{2.5})$ running time. Combining the full curve with the deterministic (min, +)-product algorithm and APSP reduction of Alman and Vassilevska Williams [1, Theorems 22 and 33–34] further gives $O(n^{2.4999})$ running time.
4. **Minimum and maximum witnesses.** For two Boolean $n \times n$ matrices, a minimum (maximum) witness for entry (i, j) is the least (greatest) index k with $A_{ik} = B_{kj} = 1$. The standard blocking algorithm [3, Section 2.1] takes $O(n^{1-r+\omega(r)} + n^{2+r})$ time. Taking $r = \frac{1}{2}$ gives $O(n^{2.5})$ time for all witnesses.
5. **Sparse matrix multiplication.** For two $n \times n$ matrices with at most m nonzero entries each, the sparse–dense decomposition of Yuster and Zwick [9, Section 3] gives $O(m^2/n^r + n^{\omega(r)})$ time for $0 \leq r \leq 1$. Thus $m \leq n^{1.25}$ permits $O(n^2)$ time.
6. **Dynamic matrix inverse and reachability.** Substituting Corollary 6.1 into van den Brand–Nanongkai–Saranurak’s tradeoff [7, Theorem 4.2] and minimizing $\max\{u + v, \omega(1, u, v) - u, \omega(v) - v\}$ over $0 \leq u \leq v \leq 1$ gives $O(n^{1.31125})$ worst-case time per entry update and query for maintaining the inverse of a nonsingular $n \times n$ matrix, after $O(n^{2.25})$ preprocessing. The associated Monte Carlo reduction gives the same bounds for directed reachability under edge updates, with $O(1)$ query time for a fixed pair [7, Corollary C.12 and Theorem C.16].

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